PERSISTENCE-ROBUST PROFILE SIEVE EMPIRICAL LIKELIHOOD ORACLE EQUIVALENCE AND EFFICIENCY TRADEOFFS

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Supplementary Material

This supplement contains complete technical details for the main manuscript. Section S1 proves the projection identities, feasible–oracle equivalence, scalar empirical-likelihood expansion, and the Wilks theorem. Section S2 derives the conditional-moment efficiency benchmark and squared-angle representation. Section S3 records additional Monte Carlo design and reproducibility details.

S1 Technical Lemmas and Proof of Theorem

Throughout the appendix, all notation is as defined in the main text. Under $H_0 : \beta = \beta_0$, write

$$\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag} = \boldsymbol{P}\boldsymbol{c}_K + \boldsymbol{r}_K + \boldsymbol{U}, \qquad \widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{U} + \boldsymbol{r}_K).$$

The equalities involving $\boldsymbol{M}_K$ are projection equalities in the empirical Hilbert space $(\mathbb{R}^T, \langle\cdot,\cdot\rangle_T)$.

Lemma S1 (Matched projection identities)**.** *Under Assumptions 2–3,*

$$\boldsymbol{P}'\boldsymbol{w}^c = \boldsymbol{0}, \qquad \|\boldsymbol{\Pi}_K \boldsymbol{U}\|_T = O_p(\sqrt{K/T}), \qquad \max_{t\le T} |(\boldsymbol{\Pi}_K\boldsymbol{U})_t| = o_p(1).$$

Proof. The first equality is exact because $\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}$ and $\boldsymbol{M}_K$ is the orthogonal projector onto $\mathrm{span}(\boldsymbol{P})^\perp$. The two bounds for $\boldsymbol{\Pi}_K\boldsymbol{U}$ are the sample projection bounds imposed in Assumption 2. In standard compact-support or truncated stationary alpha-mixing designs, they follow from the stable Gram matrix and the empirical-process rate for $T^{-1}\boldsymbol{P}'\boldsymbol{U}$. $\square$

Lemma S2 (Feasible-oracle score and variance equivalence)**.** *Under Assumptions 1–3, under $H_0 : \beta = \beta_0$,*

$$S_T := T^{-1/2}\sum_{t=1}^T \widehat{Z}_t(\beta_0) = T^{-1/2}\sum_{t=1}^T U_t w_t^c + o_p(1), \tag{S1.1}$$

$$\widehat{V}_T := T^{-1}\sum_{t=1}^T \widehat{Z}_t(\beta_0)^2 = T^{-1}\sum_{t=1}^T U_t^2 (w_t^c)^2 + o_p(1), \tag{S1.2}$$

$$\max_{t\le T} |\widehat{Z}_t(\beta_0)| = o_p(\sqrt{T}). \tag{S1.3}$$

Consequently, $\widehat{V}_T \xrightarrow{p} \eta_K^2$.

Proof. Because $\boldsymbol{M}_K$ is symmetric idempotent and $\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}$,

$$S_T = T^{-1/2}\{\boldsymbol{M}_K(\boldsymbol{U} + \boldsymbol{r}_K)\}'\boldsymbol{w}^c = T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{w}^c.$$

The approximation residual satisfies

$$|T^{-1/2}\boldsymbol{r}_K'\boldsymbol{w}^c| \le \sqrt{T}\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}^c\|_T \le \sqrt{T}\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}\|_T = o_p(1),$$

where the second inequality uses the contraction property of orthogonal projection. This proves (S1.1).

For the denominator, define the feasible-oracle distance

$$\Delta_{u,t} = \widehat{u}_t(\beta_0) - U_t, \qquad \Delta_{Z,t} = \widehat{Z}_t(\beta_0) - U_t w_t^c = \Delta_{u,t} w_t^c.$$

Here

$$\boldsymbol{\Delta}_u = -\boldsymbol{\Pi}_K \boldsymbol{U} + \boldsymbol{M}_K \boldsymbol{r}_K.$$

Assumptions 2–3 give

$$\|\boldsymbol{\Delta}_u\|_T = O_p(\sqrt{K/T} + a_K) = o_p(1).$$

Since $\max_t |w_t^c| = O_p(1)$,

$$\|\boldsymbol{\Delta}_Z\|_T^2 \le \max_t |w_t^c|^2 \|\boldsymbol{\Delta}_u\|_T^2 = o_p(1).$$

Thus the feasible score vector and the matched-projection oracle vector $(U_t w_t^c)_{t=1}^T$ are $o_p(1)$ apart in empirical distance. If $V_T^K = T^{-1}\sum_t U_t^2 (w_t^c)^2$, then

$$\widehat{V}_T - V_T^K = \frac{2}{T}\sum_{t=1}^T U_t w_t^c \Delta_{Z,t} + \|\boldsymbol{\Delta}_Z\|_T^2.$$

Cauchy–Schwarz gives

$$|\widehat{V}_T - V_T^K| \le 2(V_T^K)^{1/2}\|\boldsymbol{\Delta}_Z\|_T + \|\boldsymbol{\Delta}_Z\|_T^2 = o_p(1),$$

because $V_T^K = O_p(1)$ by Assumption 1. This proves (S1.2), and $\widehat{V}_T \to_p \eta_K^2$ follows from (3.1).

Finally, Assumption 1 gives the oracle maximum-score condition. Combining that condition with $\max_t |w_t^c| = O_p(1)$, (3.4), and the sup-norm stability of $\boldsymbol{M}_K \boldsymbol{r}_K$ proves (S1.3). □

Lemma S3 (Scalar empirical-likelihood expansion)**.** *Let $Z_{t,T}$ be scalar scores and define*

$$S_T^Z = T^{-1/2} \sum_{t=1}^{T} Z_{t,T}, \qquad \widehat{V}_T^Z = T^{-1} \sum_{t=1}^{T} Z_{t,T}^2.$$

Let $\ell_T^Z = 2\sum_{t=1}^{T} \log(1 + \widehat{\lambda} Z_{t,T})$, where $\widehat{\lambda}$ solves the scalar EL multiplier equation. Suppose the EL convex-hull condition holds, $S_T^Z = O_p(1)$, $\widehat{V}_T^Z \to_p \eta^2$ with $P(\eta^2 \geq \underline{\eta}^2) = 1$ for some $\underline{\eta} > 0$, and $\max_{t \leq T} |Z_{t,T}| = o_p(\sqrt{T})$. Then

$$\widehat{\lambda} = \frac{\sum_{t=1}^{T} Z_{t,T}}{\sum_{t=1}^{T} Z_{t,T}^2} + o_p(T^{-1/2}), \qquad \ell_T^Z = \frac{(S_T^Z)^2}{\widehat{V}_T^Z} + o_p(1).$$

Proof. The convex-hull condition gives existence of the scalar EL multiplier with probability tending to one. Since $S_T^Z = O_p(1)$ and $\widehat{V}_T^Z$ is bounded away from zero with probability tending to one, the multiplier equation implies $\widehat{\lambda} = O_p(T^{-1/2})$. The maximum-score condition then gives $\max_t |\widehat{\lambda} Z_{t,T}| = o_p(1)$. Expanding

$$0 = \sum_{t=1}^{T} \frac{Z_{t,T}}{1 + \widehat{\lambda} Z_{t,T}}$$

around zero gives

$$0 = \sum_{t=1}^{T} Z_{t,T} - \widehat{\lambda} \sum_{t=1}^{T} Z_{t,T}^2 + o_p(\sqrt{T}),$$

where the remainder is controlled by $\widehat{\lambda}^2 \sum_t |Z_{t,T}|^3 \leq \widehat{\lambda}^2 \max_t |Z_{t,T}| \sum_t Z_{t,T}^2 = o_p(\sqrt{T})$. This proves the displayed expansion for $\widehat{\lambda}$. A second-order expansion of the log likelihood gives

$$\ell_T^Z = 2\widehat{\lambda} \sum_{t=1}^{T} Z_{t,T} - \widehat{\lambda}^2 \sum_{t=1}^{T} Z_{t,T}^2 + o_p(1),$$

with remainder bounded by $|\widehat{\lambda}|^3 \max_t |Z_{t,T}| \sum_t Z_{t,T}^2 = o_p(1)$. Substituting the multiplier expansion yields $\ell_T^Z = (S_T^Z)^2 / \widehat{V}_T^Z + o_p(1)$. □

Proof of Theorem 1. By Assumption 1 and Lemma S2,

$$S_T \xrightarrow{\text{st}} MN(0, \eta_K^2), \qquad \widehat{V}_T \xrightarrow{p} \eta_K^2.$$

Thus $S_T = O_p(1)$, and self-normalization gives $S_T^2 / \widehat{V}_T \xrightarrow{d} \chi_1^2$. Applying Lemma S3 to $Z_{t,T} = \widehat{Z}_t(\beta_0)$ gives

$$\ell_T(\beta_0) = \frac{S_T^2}{\widehat{V}_T} + o_p(1).$$

The result follows by Slutsky's theorem. □

S2 Efficiency Benchmark Derivations

Derivation of Proposition 1. Suppress the time subscript. The conditional-mean score direction for β is qXU. A nuisance perturbation $m_\eta(W) =$

$m(W)+\eta h(W)$ has score direction $qh(W)U$. The least favorable projection therefore chooses $\mu_q(W)$ so that

$$E\left[\{qXU - q\mu_q(W)U\}qh(W)U\right] = E\left[q\{X - \mu_q(W)\}h(W)\right] = 0$$

for every square-integrable $h(W)$. This normal equation gives

$$\mu_q(W) = \frac{E(qX \mid W)}{E(q \mid W)}.$$

The residual score direction is $q\{X - \mu_q(W)\}U = qVU$. Its variance is

$$E\{(qVU)^2\} = E(q^2V^2U^2) = E(qV^2),$$

which is the conditional-mean efficient information. The efficient influence function is the inverse information times the efficient score. □

Derivation of Proposition 2. The sample moment based on g has derivative $-E(gX)$ with respect to β and long-run one-period variance $E(g^2U^2) = E(\sigma^2g^2)$ under the maintained martingale-difference setup. Because g is orthogonal to every function of W,

$$E(gX) = E\{g[X - \mu_q(W)]\} = E(gV).$$

Thus

$$\mathrm{AVAR}(g) = \frac{E(\sigma^2g^2)}{\{E(gV)\}^2}.$$

With $g^\star = qV$ and $\langle a, b\rangle_{\sigma^2} = E(\sigma^2ab)$,

$$\langle g, g^\star\rangle_{\sigma^2} = E(gV), \qquad \|g^\star\|^2_{\sigma^2} = E(V^2/\sigma^2) = \mathcal{I}_{\mathrm{eff}}.$$

Therefore

$$\mathrm{RE}(g) = \frac{\langle g, g^\star \rangle_{\sigma^2}^2}{\|g\|_{\sigma^2}^2 \|g^\star\|_{\sigma^2}^2} = \cos_{\sigma^2}^2 \angle(g, g^\star) \leq 1.$$

The equality condition in Cauchy–Schwarz is exact collinearity, so equality holds if and only if $g \propto g^\star = qV$. □

Derivation of Proposition 3. The first statement is the equality case of Proposition 2: using $g_t^\star = q_t V_t$ makes the weighted angle with the efficient direction equal to zero, so $\mathrm{RE}(g^\star) = 1$. In the homoskedastic submodel, q_t is a common constant and can be dropped from a scalar EL moment. The efficient instrument is therefore $X_{t-1} - E(X_{t-1} \mid W_{t-1})$, and the sieve analogue is $\boldsymbol{M}_K \boldsymbol{X}_{lag}$. The displayed formula for $\widehat{\beta}_{\text{eff}}$ is the root of the corresponding residualized sample moment.

For the persistent triangular-array statement, write the oracle efficient score as $\xi_{t,T} = q_t V_{t,T} U_t$. If the scaled oracle CLT, denominator consistency, maximum-score condition, and feasible–oracle replacement hold at the information scale r_T, then Lemma S3 applies to the normalized score $(\sqrt{T}/r_T)\xi_{t,T}$. Multiplying all EL moments by the common nonzero factor $\sqrt{T}/r_T$ leaves the EL constraint and likelihood ratio unchanged, so the normalization only records the proof scale. The resulting efficient-weight statistic therefore attains the conditional-moment information bound associated with J. □

S3 Monte Carlo Design and Reproducibility Details

The headline simulation design uses

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t, \qquad X_t = \rho_T X_{t-1} + V_t, \qquad W_t = a_W W_{t-1} + \eta_t,$$

with $a_W = 0.5$, W_t initialized from its stationary Gaussian distribution, and

$$\begin{pmatrix} U_t \\ V_t \\ \eta_t \end{pmatrix} \sim N\left(0, \begin{pmatrix} 1 & \kappa & 0 \\ \kappa & 1 & \xi \\ 0 & \xi & 1 \end{pmatrix}\right).$$

The headline Wilks design sets $\kappa = 0$, $\xi = 0.3$, $m(w) = 0.5\sin(w) + 0.3(w^2 - 1)$, $K = 6$, and initializes the persistent predictor from $X_0 = 0$ at the sample start. The nuisance process receives a burn-in; the unit-root predictor is not given a random-walk burn-in because such a burn-in is not a stationary initialization device.

The main bounded weight is $w_b(x) = \tanh(x/b)$ with $b = 8$, which keeps the matched-projection unit-root score nondegenerate in the $T = 250$ design. The frontier experiment uses $b \in \{0.25, 0.5, 1, 2, 4, 8\}$. The headline size table in the main paper is generated by `configs/mc/size_main.yaml`, and the frontier figure is generated by `configs/mc/frontier_main.yaml`. Each run writes the raw replication file, config hash, manifest, environment record, deterministic summary tables, and figures.

A stress design with $\kappa = 0.5$ is kept separate from the headline Wilks design. At $T = 250$ and $b = 8$, the unit-root oracle and profile rejection frequencies are both 0.113 in the current simulation output. This stress result is used only to show that the high-level oracle Wilks conditions are substantive under contemporaneous endogeneity and unit-root persistence; it is not a sieve-profiling failure because the feasible statistic continues to track its matched-projection oracle counterpart.